Q3(a)	Find the minimal expression and realize it with basic gates. $F(A,B,C,D)=\sum m(0,1,3,5,7,9,10,14,15)+d(2,4)$	2	CO3	K5	
Q3(b)	Write down the truth table of full adder, find the minimal expression of ~~Difference, borrow~~ sum, carry and realize these expressions with basic gates.	3	CO3	K5	
Q3(c)	Draw the circuit diagram of JK and T Flip Flop. Write expression for output and draw its characteristic table.	5	CO3	K4	

3(a)	Find the minimal expression and realize it with basic gates. $F(A,B,C,D)=\sum m\,(0,1,3,5,7,9,10,14,15) + d(2,4)$	5	CO3	K5, K6
3(b)	Explain S-R flip flop with suitable diagram and show the Truth Table, Characteristic Table, Characteristic Equation and Excitation Table.	5	CO3	K3, K4

3(a)	Find the minimal expression and realize it with basic gates. $F(A,B,C,D)=\sum m\,(0,1,3,5,7,9,10,14,15) + d(2,4)$	5	CO3	K5, K6
3(b)	Explain S-R flip flop with suitable diagram and show the Truth Table, Characteristic Table, Characteristic Equation and Excitation Table.	5	CO3	K3, K4

3(a)	Explain the working of J-K Flip Flop with circuit diagram. Write its truth table. Also show Excitation table and Characteristic table.	3	3
(b)	Explain the Full Adder with its circuit diagram. Also show the expression for sum and carry output.	3	3
(c)	Solve the following using K-Map- $F(A,B,C,D) = \sum (0,3,4,7,8) + d\,(10,11,12,13,14,15)$ Also realize the output using logic gates	4	3